

Sinha Afroz

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Research Interests

- Extended Reality (XR), Virtual Reality (VR), and Mixed Reality (MR)
- Generative Artificial Intelligence and Human–AI Interaction
- Mental Health Technologies and Digital Therapeutics
- Medical Imaging and 3D Visualization
- Human-Computer Interaction (HCI)

Education

PhD in Computer Science Fall 2025–Present
Kennesaw State University (KSU), Kennesaw, Georgia, USA
Grade Point Average (GPA): 4.00/4.00
Advisor: Dr. Lei Zhang

BSc. in Computer Science and Engineering 2017–2022
United International University (UIU), Dhaka, Bangladesh
Cumulative Grade Point Average (CGPA): 3.89/4.00 (Magna Cum Laude)

Research Experience

Generative VR for Mental Health – NSF-Funded Project 2025–Present
Kennesaw State University, Advisor: Dr. Lei Zhang

- Designing a human-in-the-loop immersive generative virtual reality system that creates personalized therapeutic environments from users’ emotional descriptions using large language models.
- Developing an emotion-aware workflow that allows users to review, edit, and select AI-generated virtual environments while incorporating safety filtering and human oversight.
- Implementing the system in Unity for Meta Quest using OpenAI models, voice interaction, and 360° environment generation.
- Conducting pilot studies to evaluate usability, user experience, and the potential of AI-assisted immersive environments for imagery-based psychotherapeutic practices.

Professional Employment

Junior Software Engineer Jan 2023 – Dec 2023
Dhaka Popular Travel Ltd., Software Development Team, Dhaka, Bangladesh

- Built B2B airline ticketing platform serving international clients using Laravel, Vue.js, Tailwind CSS, and MySQL; maintained 99%+ uptime through Docker-based deployment on Linux.
- Automated data extraction pipelines with Python, producing and publishing a curated Airports & Airlines dataset ([Kaggle](#)) adopted by the community.
- Streamlined development workflows with Git/GitHub, reducing merge conflicts and improving team release cadence.

Publications

Conference Publications

- **Sinha Afroz**, Wendy G. Lichtenthal, Erica C. Kaye, Lei Zhang. *A Human-in-the-Loop Immersive Generative Virtual Reality System for Imagery-Based Psychotherapeutic Practices*. Accepted to the ISMAR 2026 Workshop on Mixed/Augmented Reality for Mental Health (MARMH'26), 2026.(Accepted)
- **Sinha Afroz**, Devon Haynes, Reece Freeman, Mason Prather, Feiyu Lu, Brooke Zhao, Joy Li, Sungchul Jung, Lei Zhang. *Beyond Slices: A Narrative-Driven, Multi-User, Multi-Modal Virtual Reality System for Medical Imaging Presentation*. Proceedings of the 2026 IEEE Conference on Virtual Reality and 3D User Interfaces Workshops (VRW), 3DUI Contest Demo, 2026.
DOI: [10.1109/VRW62533.2025.11489916](https://doi.org/10.1109/VRW62533.2025.11489916)
- Md. Ashif Mahmud Joy, Md. Fuad Hasan Khan Chowdhury, **Sinha Afroz**, et al. *Real-Time Face Recognition with Mask using Deep Convolutional Neural Network*. Proceedings of the 4th International Conference on Computing, Networks, and Internet of Things (CNIOT 2023), Xiamen, China, pp. 457–461, 2023.
DOI: [10.1145/3603781.3603863](https://doi.org/10.1145/3603781.3603863)

Journal Articles

- **Sinha Afroz**, Lei Zhang. *Immersive Extended Reality (XR) in Oncology: A Scoping Review of Virtual Reality (VR) and Mixed Reality (MR) Applications for 3D Solid Tumor Visualization*. Frontiers in Virtual Reality, 2026.
DOI: [10.3389/frvir.2026.1780348](https://doi.org/10.3389/frvir.2026.1780348)

Teaching & Mentoring

Undergraduate Research Mentor

Summer 2026

Kennesaw State University – DreamSpace Lab

- Mentored an undergraduate student in Unity and Virtual Reality development.
- Provided technical guidance on VR application development, project implementation, debugging, and Git/GitHub workflows.
- Reviewed project progress and deliverables and provided feedback during regular mentoring meetings.

Secondary Instructor – Vertically Integrated Projects (VIP)

Fall 2026

Kennesaw State University

- Serve as a secondary instructor supporting undergraduate students working on Unity and Virtual Reality research projects.
- Mentor students by defining project milestones and deliverables, reviewing progress, and troubleshooting technical issues.
- Provide technical guidance on Unity, VR development, OpenXR, multiplayer development, and Git/GitHub workflows.
- Coordinate student assignments, project progress, and weekly meetings with the faculty instructor.

Presentations & Demos

3DUI Contest Demo – *Beyond Slices: A Narrative-Driven, Multi-User, Multi-Modal VR System for Medical Imaging Presentation*

IEEE VR 2026

Skills

Programming Languages: Python, Java, C, C#

XR/VR Development: Unity, XR Interaction Toolkit, Netcode for GameObjects, Meta Quest (2/3), OpenXR

Medical Imaging & 3D: 3D Slicer

ML/AI Libraries: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas

Web & Databases: HTML, CSS, JavaScript, Vue.js, Laravel, MySQL

Tools & DevOps: VS Code, Jupyter, Docker, Anaconda, Git/GitHub

Others: L^AT_EX, Microsoft Office

Selected Projects

Beyond Slices: Multi-User VR System for Medical Imaging Published, IEEE VR 2026

- Collaborative VR system for surgeon-family communication in pediatric oncology; built with Unity, XR Interaction Toolkit, Netcode for GameObjects, deployed on Meta Quest 3.

Real-Time Face Recognition With and Without Mask Published, CNIOT'23

- Deep CNN-based face recognition system operating under masked and unmasked conditions; accompanied by a compact hardware demonstration prototype.

Dhaka RentVision: Smart Rental Price Prediction

- Trained and tuned ML regression models with GridSearchCV; deployed predictions via a Flask web interface.

DNA-Binding Protein Identification

- Binary classification of DNA-binding proteins using DNN (MLP) and SVM; evaluated against standard bioinformatics benchmarks.

Water Cleaning Boat 1st Prize, UIU CSE Project Show 2019

- Autonomous surface waste collection boat controlled via Raspberry Pi and Arduino Uno.

Other projects: [Online Tailor Shop](#) (Laravel/Vue.js), [Bus Ticket Reservation](#) (Java), [Save The Butterfly](#) (OpenGL game), Popular Travel B2B Italy Website (Laravel/Vue.js/MySQL).

Honors & Awards

Dean's 4.0 Club Fall 2025, Spring 2026
Kennesaw State University – recognition for perfect GPA in first semester of PhD program.

Magna Cum Laude 2024
United International University – awarded at convocation for consistent academic excellence (CGPA 3.89/4.00).

Merit Scholarship 2017–2022
United International University – full scholarship maintained across all academic years based on performance.

1st Prize – Water Cleaning Bot 2019
UIU CSE Project Show – awarded for developing an autonomous surface waste management system for water bodies.

Professional Activities & Service

Association for Computing Machinery (ACM) – Complimentary Professional Member	2023–2024
Former Member, UIU Robotics Club	2018–2019
Former Member, UIU Computer Club	2017–2018
Participant, Bangladesh Math Olympiad	2012